THE UNIVERSITY OF DANANG
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FACULTY OF ADVANCED SCIENCE AND TECHNOLOGY

CAPSTONE PROJECT REPORT
M1 REPORT

DESIGN AND IMPLEMENT OF AN INTELLIGENT EDGE AIOT WEARABLE SYSTEM FOR REAL-TIME SAFETY MONITORING OF ELDERLY PEOPLE

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1. Introduction

1.1. Background and Problem Statement

The rapid expansion of the global elderly population has generated an increasing demand for advanced healthcare monitoring systems that facilitate independent living while ensuring safety and timely medical intervention [1]. According to the World Health Organization, falls represent one of the primary causes of injury-related morbidity and mortality among older adults worldwide, often leading to a catastrophic decline in quality of life [1]. Elderly individuals, particularly those living alone, are especially susceptible to undetected fall incidents and acute physiological deterioration, including abnormal heart rate, decreased blood oxygen saturation (SpO_2), and irregular activity patterns [3]. In such scenarios, delayed emergency response substantially elevates the risk of severe trauma, prolonged hospitalization, and long-term functional impairment [1], [5].

Beyond fall-related injuries, cardiovascular diseases (CVDs) and stroke continue to rank among the leading causes of mortality and disability in the elderly population globally [6]. Statistics from the American Heart Association indicate that cardiovascular-related complications are responsible for millions of deaths annually, many of which occur in home settings without immediate supervision [7]. Clinical evidence suggests that early abnormalities in physiological parameters—such as heart rate variability, cardiac arrhythmias, and hypoxemia—are strongly associated with an increased likelihood of stroke, cardiac arrest, and respiratory failure [2], [8]. Therefore, the current healthcare landscape faces a critical challenge: providing a proactive safety net that can autonomously bridge the gap between home-based living and emergency medical services.

1.2. Technological Evolution: Transitioning to Edge AIoT

Recent advancements in wearable sensing and IoT infrastructures have enabled continuous physiological data acquisition beyond traditional settings [3], [6]. However, existing healthcare monitoring solutions remain heavily dependent on cloud-centric architectures for data processing. These cloud-reliant systems face critical limitations: high Network Latency in emergencies, which compromises rescue effectiveness [4]; Privacy Vulnerabilities due to transmitting sensitive medical data over public networks [7]; and high Power Consumption, which renders 24/7 wearable operation impractical [4].

Edge computing addresses these drawbacks by relocating computational intelligence closer to the wearable device, thereby minimizing latency and enhancing real-time decision-making capabilities [4]. While machine learning has significantly improved fall detection accuracy, deploying complex models on resource-constrained embedded systems remains a major technical challenge, largely due to the inherent trade-off between accuracy and computational efficiency [5], [8]. Consequently, there is a compelling need to design and implement an Intelligent Edge AIoT wearable system capable of performing real-time health index computation, continuous heart rate and SpO_2 monitoring, and accurate fall detection directly at the device level. Such a system is essential to reduce response latency, strengthen data privacy protection, and significantly enhance the effectiveness of emergency interventions in elderly safety monitoring applications.

1.3. Existing Wearable Safety Monitoring Solutions and Their Limitations

Recent technological advancements have led to the proliferation of commercial wearable devices designed for health, fitness, and medical alert purposes. To position our proposed Edge AIoT system effectively, a comparative analysis of leading products in the market is essential. This benchmarking focuses on devices that address, directly or indirectly, the core needs of elderly safety monitoring: continuous vital sign measurement, fall detection, and robust alert systems. The analysis highlights key product differentiators and, more critically, the gaps in functionality and architecture that our project aims to bridge.

Table 1: Existing Solution Benchmarking

Feature / Criteria	**WHOOP 4.0/5.0**	**Medical Guardian (MGMini)**	**Oura Ring Gen 3/4**	**Angel Watch (Series R)**
				
Market Segment	Elite Fitness & Athletes	Medical Alert & Rescue	Wellness & Lifestyle	Senior Smartwatch
Form Factor	Screenless Wrist/Bicep Band	Clip-on or Pendant	Ultra-small Smart Ring	Smartwatch (With Screen)
HR & SpO2 Sensors	YES	NO	YES	YES
Fall Detection	NO	YES	NO	YES
Alert System	NO	YES	NO	YES
Connectivity	**BLE 5.2**	**4G LTE + WiFi + GPS**	**BLE 5.2**	**4G LTE + WiFi + GPS**
Total Year 1 Cost	**~$199.00**	**~$750**	**~$421**	**~$240**

1.4. Marketing and Engineering requirements:

The design and implementation of the Edge AIoT wearable system are guided by a set of well-defined requirements, translating market demands for elderly safety and usability into rigorous engineering specifications. The marketing requirements (A-H) are defined as follows:

- **A:** Effectiveness and Reliability
- **B:** Timely Response
- **C:** Health Monitoring
- **D:** Optimal Battery Life
- **E:** Ease of Use
- **F:** User-Friendly Design
- **G:** Edge AI Capability
- **H:** Competitive Cost

These requirements are summarized below, mapping high-level marketing goals to measurable engineering metrics and their respective justifications.

Table 2: Engineering requirement for product

Marketing Requirement	Engineering Requirement	Justification
A	Algorithm Accuracy: Fall Detection Accuracy $\geq$ 95% on the selected dataset.	Ensures maximum user safety and minimizes false alarms.
B, G	Processing Latency: The time from event detection (fall/anomaly) to alert transmission over the network must be $<$ 2 seconds.	Ensures timely emergency response (B) and highlights the speed advantage of Edge AI (G).
C	Sensor Accuracy: Measure HR $\pm$ 2 BPM and SpO_2 $\pm$ 3%, meeting basic health monitoring device standards.	Provides reliable physiological data for proactive health monitoring (C).

D, G	Power Efficiency: The system's average power consumption must be optimized to ensure continuous operation for ≥ 5 days.	Ensures 24/7 monitoring availability (D). Optimized Edge AI models help reduce power consumption (G).
E	Connectivity & Synchronization: Ensure stable BLE 5.0 connection and achieve 100% automatic data synchronization with the mobile application (within range).	Simplifies usability, enabling caregivers to easily set up and monitor data without manual intervention (E).
F, H	Physical Specifications: The device must weigh ≤ 70 g, use skin-friendly materials, and meet IP67 water/dust resistance standards.	Improves wearing comfort and encourages continuous usage (F). Material choice supports cost targets (H).
B, D, G	Edge AI Optimization: AI model size must be ≤ 500 KB and inference speed on MCU must be ≥ 10 FPS (10 ensures latency < 2 seconds).	Enables fast processing to meet latency requirements (B, G) and reduces CPU load to save energy (D, G).
H	BOM Cost Management: Total Bill of Materials (BOM) cost per unit must be ≤ 2,500,000 VND.	Ensures commercial viability and competitive retail pricing against imported products (H).

2. Preliminary solution

2.1. System Architecture Block Diagram

The proposed Intelligent Edge AIoT Wearable System is designed to enhance the safety and promote independent living for the elderly by directly addressing the critical issues of latency, privacy, and power consumption inherent in conventional cloud-based monitoring solutions [4]. The architecture shifts complex analytical capabilities closer to the user by autonomously processing multi-parameter data from dedicated motion-tracking and physiological vital sign sensors in response to raw sensor input [3]. The core mechanism involves performing real-time inference directly on the

Microcontroller Unit (MCU) to eliminate debilitating network latency [4]. This Edge AI approach enables continuous monitoring and rapid identification of cardiovascular anomalies for at-risk populations [4]. By utilizing an optimized, lightweight AI model, the system accurately assesses the user's condition and detects high-risk events such as falls or acute hypoxemia, offering a more precise and adaptive solution compared to generic medical alert devices [5]. Ultimately, this architecture improves the quality and effectiveness of emergency response by ensuring continuous physiological monitoring and enabling preemptive alerts.

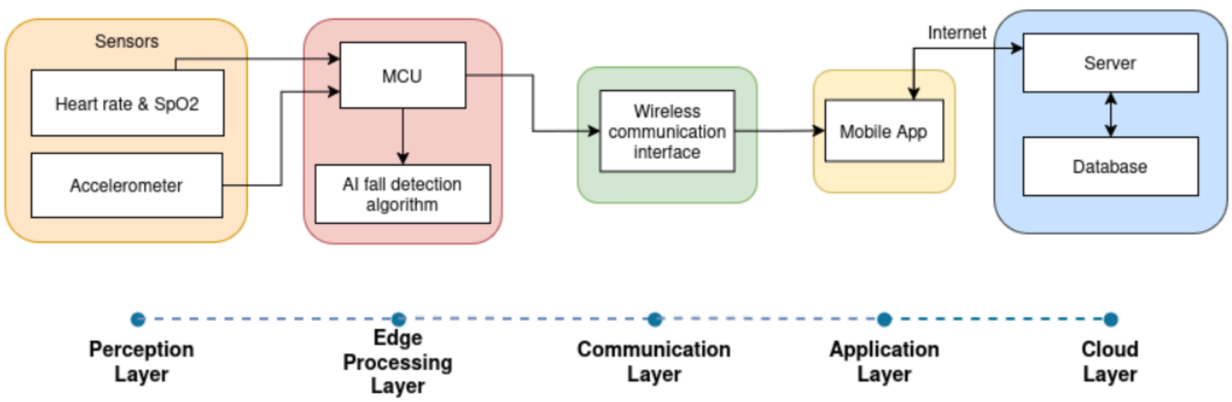

Figure 1: Proposed System Architecture

2.2. Preliminary Technical Solution Description

2.2.1. Data Acquisition Module

The Data Acquisition Module is responsible for continuously collecting both motion and physiological signals from the user, providing the fundamental input for subsequent edge processing and decision-making. Motion data is captured by a wearable Inertial Measurement Unit (IMU), which integrates a three-axis accelerometer and gyroscope to capture dynamic body movements. This raw IMU data is preprocessed (via noise filtering and smoothing) before being forwarded to the Edge AI pipeline for identifying abnormal movement patterns and potential fall events. Concurrently, physiological data, specifically Heart Rate (HR) and Blood Oxygen Saturation (SpO_2), is acquired via an optical sensing mechanism based on Photoplethysmography (PPG). The raw PPG signals undergo signal conditioning to obtain stable and reliable measurements. By integrating these two complementary sources (motion sensing and physiological monitoring) into a multi-modal data acquisition framework, the module significantly enhances the robustness and reliability of real-time fall detection and overall safety monitoring in real-world elderly care scenarios.

2.2.2. Design Considerations and Technology Selection

To ensure system performance, energy efficiency, and feasibility for wearable deployment, key components and data sources are selected based on a structured benchmarking process. Each candidate is evaluated using criteria such as power consumption, integration level, computational capability, and suitability for real-world applications.

Table 3: Benchmarking of Wearable Motion Datasets for Fall Detection

Dataset	Sensor	Samples	Subjects	Fall types	ADL diversity	Sampling rate	Expected accuracy
WEDA-FALL[27]	wrist accel + gyro	~776 events	~20	8	11	50Hz	95–98%
HIFD (HR-IMU)[28]	wrist accel + gyro	~1200 events	21	6	9	50Hz	93–97%
UP-Fall (wrist subset)[29]	wrist accel + gyro	~1100 events	17	5	6	50Hz	92–96%
UMA-Fall (wrist subset)[30]	wrist IMU	~2000 events	17	8	12	20–200Hz	93–97%
Capture-24 [31]	wrist accelerometer	3883 hours	151	0	very high	100Hz	N/A (ADL only)
UCI Wrist ADL dataset [32]	wrist accelerometer	~3000 sequences	16	0	moderate	32Hz	N/A

The dataset comparison indicates that wrist-based datasets provide the most consistent and relevant motion patterns for wearable applications, as they closely match the actual device placement. Among the evaluated datasets, WEDA-FALL and HIFD are selected as the primary datasets due to their balanced number of subjects, diverse fall types, and stable sampling rate (50 Hz), which are suitable for real-time embedded processing. In addition, UMA-Fall and UP-Fall are incorporated to further increase the diversity of motion patterns and improve model generalization across different users and scenarios.

Datasets such as Capture-24 and UCI Wrist ADL are not selected for fall detection training, as they mainly contain daily activities without fall events, making them more suitable for non-fall activity modeling rather than classification.

Therefore, a multi-dataset strategy is adopted, combining selected wrist-based fall datasets to achieve a balance between data diversity, model accuracy, and practical deployment constraints.

Table 4: Sensors benchmarking (Heart rate & SpO_2*)*

Criteria	AFE4404 [33]	ADPD144RI [34]	AS7030B [35]	MAX30102 [36]
Heart Rate & SpO_2	✅	✅	✅	✅
Integration level	AFE only	Integrated PD + AFE	Integrated optical biosensor	LED + PD + AFE integrated
Power consumption	~1–2 mA	~300–400 µA	~150–300 µA	~600 µA typical
Wearable suitability	Moderate	High	High	Very high
Communication	SPI	I²C / SPI	I²C	I²C

The comparison indicates that while some sensors offer lower power consumption, they require more complex external circuitry or optical configurations. The MAX30102 provides a highly integrated solution, simplifying hardware design while maintaining reliable signal acquisition. Therefore, it is selected as the physiological sensor.

Table 5: Sensors benchmarking for fall detection

Criteria	ADXL345 [37]	MPU6050 [38]	BMI160 [39]	LSM6DS3 [40]
3-axis Accelerometer	✅	✅	✅	✅
3-axis Gyroscope	❌	✅	✅	✅
Accelerometer Range	±16g	±16g	±16g	±16g
Accelerometer Noise Density	~220 µg/√Hz	~400 µg/√Hz	~150 µg/√Hz	~90 µg/√Hz
Gyroscope Noise Density	❌	~0.01 °/s/√Hz	~0.007 °/s/√Hz	~0.004 °/s/√Hz
Power Consumption (Typical)	~140 µA	~3.9 mA	~950 µA	~0.9 mA
Package Size	3 × 5 × 1 mm	4 × 4 × 0.9 mm	2.5 × 3 × 0.8 mm	2.5 × 3 × 0.83 mm

The results show that sensors lacking gyroscope functionality are insufficient for fall detection, while some alternatives consume excessive power. The BMI160 provides a balanced trade-off between sensing capability, noise performance, and energy efficiency, making it suitable for continuous wearable monitoring. Therefore, BMI160 is selected.

Table 6: Main MCU benchmarking

Criteria	ESP32-C3 Mini [41]	ESP32-S3 Mini [42]	STM32L432KC [43]
CPU	RISC-V 160 MHz	Dual-core 240 MHz	Cortex-M4F 80 MHz
RAM	~400 KB SRAM	512 KB + 8 KB	64 KB SRAM

		RTC	
TinyML Support	✅	✅	✅
WiFi	✅	✅	❌
BLE	BLE 5.0	BLE 5.0	❌
Operating Current	~80–100 mA (WiFi) / ~20–30 mA (BLE)	~100–120 mA (WiFi) / ~25-40 mA (BLE)	~8–12 mA
Deep Sleep Current	~5 µA	~5 µA	~1 µA
Suitability for Wearable AI	Good	Very suitable	Limited connectivity

From the comparison, low-power microcontrollers are not sufficient to support Edge AI processing, while other options increase system complexity due to missing integrated connectivity. The ESP32-S3 offers strong computational capability, sufficient memory, and built-in wireless communication, making it the most suitable choice. Therefore, ESP32-S3 Mini is selected as the MCU.

2.2.3. Edge Processing & Decision-Making Module

The edge processing module performs real-time analysis and decision-making directly on the device, reducing latency and improving data privacy. To achieve an optimal balance between accuracy and computational efficiency, different fall detection approaches are evaluated.

Table 7: AI Algorithm for fall detection benchmarking

Approach	Algorithm	Memory Usage (RAM)	Model Size (Parameters)	Ability to Distinguish Real vs False Falls (F1)	Training Data Needed (samples)
Threshold-based	Rule-based acceleration threshold	~1–2 KB	0	~0.70–0.80	<200

Feature ML	Random Forest	~10–50 KB	~1k–5k nodes	~0.90–0.96	500–2000
Feature ML	XGBoost / Gradient Boost	~30–120 KB	~5k–20k nodes	~0.92–0.97	1000–5000
Neural Network	MLP (Dense)	~20–80 KB	~10k–40k	~0.90–0.95	1000–5000
Temporal ML	LSTM	~150–500 KB	~100k–300k	~0.94–0.98	5000–20000
Temporal ML	CNN–LSTM	~300 KB – 1 MB	~200k–600k	~0.95–0.99	10000–30000
Tiny Deep Learning	1D CNN (TinyCNN)	~30–120 KB	~20k–60k	~0.95–0.99	2000–10000
Advanced TinyML	ConvLSTM / TinyFallNet	~80–200 KB	~60k–150k	~0.96–0.99	5000–15000
Hybrid Edge AI	Threshold + CNN	~30–120 KB	~20k–60k	~0.97–0.99	2000–10000

The comparison highlights a clear trade-off between accuracy and resource requirements. Lightweight methods lack reliability, while more complex models exceed the limitations of embedded systems. The Hybrid Edge AI approach achieves a strong balance by combining efficient computation with high detection performance. Therefore, this approach is selected for real-time fall detection.

2.2.4. Communication and Alert Module

The communication module is responsible for transmitting processed data and emergency alerts from the wearable device to external systems. Different wireless protocols are evaluated based on power consumption, communication range, and compatibility with mobile devices.

Table 8: Wireless protocol benchmarking

Protocol	Power Consumption	Range	Data Rate	Smartphone Support

Bluetooth Low Energy (BLE)	Very Low	~10–50 m	~1 Mbps	Native support
Wi-Fi	High	~50 m	Up to hundreds Mbps	Native support
Zigbee	Low	~10–100 m	~250 kbps	Limited
LoRa	Very Low	Up to several km	Very Low	Not supported
Cellular (4G / NB-IoT)	High	Very Long	Medium	Not required

The results show that high-data-rate protocols consume excessive power, while long-range technologies lack practical compatibility with smartphones. Bluetooth Low Energy provides the most suitable balance between low power consumption, sufficient range, and seamless integration with mobile devices. Therefore, BLE is selected as the primary communication protocol.

3. Expected results

The project is expected to deliver a fully functional Wearable Edge AIoT System Prototype, integrating key components—the BMI160 motion sensor, MAX30102 physiological sensor, and ESP32-S3 Mini microcontroller—within a **lightweight (≤ 70 g)** and **durable (IP67)** physical design suitable for **continuous 24/7** use. Central to this achievement is the successful deployment of an Optimized Edge AI Model **(size ≤ 500 KB)** directly onto the MCU, which is projected to achieve a robust fall detection **accuracy of ≥ 95%** and provide high-fidelity physiological monitoring **(HR ± 2 BPM, SpO_2 ± 3%)**.

This system is targeted to meet stringent operational performance metrics, including an ultra-low processing **latency of ≤ 2 seconds** from detection to alert transmission and energy efficiency supporting continuous **operation for ≥ 5 days**. These results will be accessed via a dedicated Caregiver Mobile Application that ensures stable **BLE 5.0** connectivity and instant emergency alerts, all while maintaining commercial viability with a total Bill of Materials (BOM) **cost ≤ 2,500,000 VND**.

4. Evaluation method

This evaluation phase aims to assess the effectiveness, reliability, and commercial viability of the proposed system through comprehensive testing methodologies. The evaluation is primarily focused on verifying compliance with the defined Engineering Requirements.

4.1. Evaluation of Algorithm Accuracy and Processing Latency.

This evaluation focuses on validating the system's core safety functionality and real-time performance. The accuracy of the fall detection algorithm will be rigorously assessed using either real-world or simulated fall datasets. Standard performance metrics, including Accuracy, Sensitivity, Specificity, and F1-score, will be employed to ensure that the model achieves a minimum performance threshold of ≥ 95% (Requirement A).

In parallel, to demonstrate the advantages of the Edge AI architecture, processing latency is considered a critical metric. It will be carefully measured under controlled experimental conditions. Specifically, the total latency—from the moment a fall event is captured by the sensors to the successful transmission of an alert packet via BLE—must remain below 2 seconds (< 2 s), satisfying Requirements B and G.

4.2. Evaluation of Physiological Sensor Accuracy and Power Efficiency.

The objective of this evaluation is to verify the reliability of medical data and the system's capability for continuous operation. The accuracy of physiological measurements will be validated by comparing heart rate (HR, ±2 BPM) and blood oxygen saturation (SpO_2, ±3%) readings from the wearable device against a certified medical-grade reference device (gold standard) under controlled test conditions. The Mean Absolute Error will be calculated to ensure compliance with the specified accuracy requirements. For continuous 24/7 monitoring, power efficiency will be evaluated by measuring the average current consumption of the device across different operating modes (e.g., Active and Sleep). Based on these measurements, the total battery life will be estimated and must achieve a minimum continuous operation time of ≥ 5 days, satisfying Requirements D and G.

4.3. Evaluation of Edge AI Optimization and Bill of Materials (BOM)

This section evaluates two key factors that determine the system's technological differentiation and market competitiveness. First, AI model optimization will be validated using tools such as TensorFlow Lite Micro benchmarking. The model size must not exceed 500 KB, and the inference speed on the MCU must be sufficiently fast to support the end-to-end latency requirement of ≤ 2 seconds (Requirement G). Second, from an economic perspective, a detailed Bill of Materials (BOM) will be developed to estimate production costs. The total BOM cost must not exceed 2,500,000 VND, ensuring affordability and commercial feasibility in the target market .

4.4. Evaluation of Usability and Physical Specifications

Usability and physical design are critical factors influencing user acceptance, particularly for elderly users. User Acceptance Testing (UAT) will be conducted to collect feedback on wearing comfort, including device weight (≤ 70 g) and the use of skin-friendly materials. Additionally, the usability of the mobile application will be evaluated in terms of intuitiveness, ease of setup, and daily operation. From a system perspective, the stability of the Bluetooth Low Energy (BLE 5.0) connection and the reliability of automatic data synchronization will be tested in real-world scenarios (Requirement E).

Furthermore, the device's resistance to water and dust will be validated according to the IP67 standard (Requirement F).

5. Plan of implementation

The table below (Table 9) illustrates the specific tasks assigned to each team member. This matrix ensures that every component of the system—from AI model development to hardware assembly—is accounted for, promoting accountability and specialized focus throughout the project lifecycle.

Table 9: The tasks are included in each stage

Task #	Task Details	Timeline	Person in charge	
			Phuoc Dien	Van Tien
1	Research elderly health issues and relevant safety statistics	02/03 - 08/03	✔	✔
2	Analyze user personas and define system requirements	09/03 - 15/03		✔
3	Benchmark commercial products and study technical solutions	16/03 - 22/03	✔	✔
4	Design the preliminary system architecture	23/03 - 28/03	✔	✔
5	Acquire and preprocess datasets for AI model development	29/03 - 10/04	✔	
6	Design and train the core AI model for fall detection	11/04 - 20/04	✔	
7	Develop the mobile application UI/UX and backend logic	29/03 - 20/04	✔	✔

8	Optimize the AI model for efficient edge deployment	21/04 - 30/04	✔	
9	Develop device firmware and sensor acquisition modules	01/05 - 12/05	✔	✔
10	Design PCB schematic and assemble SMD hardware components	13/05 - 25/05		✔
11	Integrate software, hardware, and wireless communication	26/05 - 31/05	✔	✔
12	Design the 3D enclosure	01/06 - 10/06	✔	
13	Evaluate system performance and overall reliability metrics	10/06 - 20/06	✔	✔

The Gantt Chart in Figure 2 illustrates the project's schedule and task dependencies. It highlights a strategic progression from foundational research and architectural design to parallel development of the AI model and mobile application, concluding with a comprehensive integration and system evaluation phase.

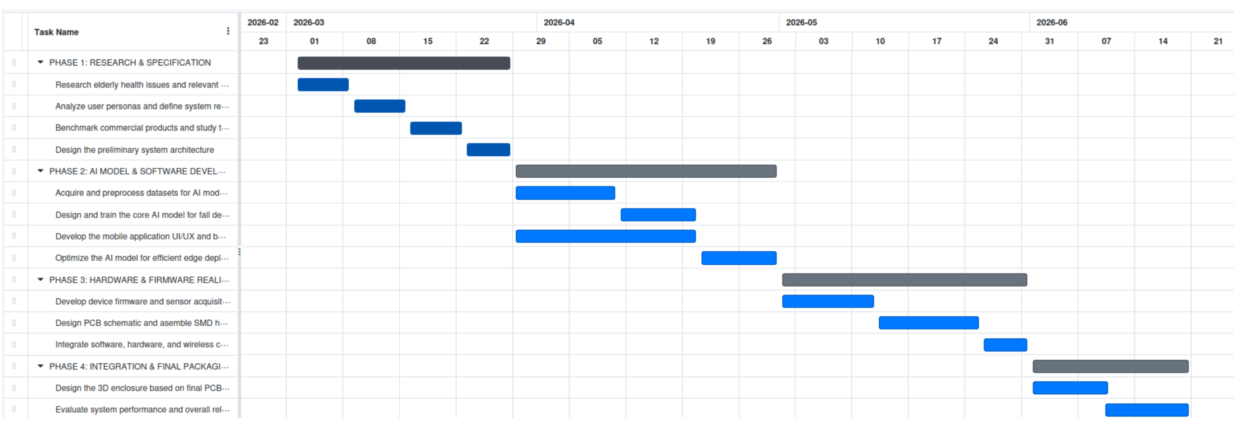

Figure 2: Gantt Chart of Task Assignment

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